POST-HURRICANE OPPORTUNITIES

Although coping with hurricanes is difficult, it does offer the opportunity to make homes more energy efficient and comfortable.

BY DANNY PARKER


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When Florida was slammed by a record four hurricanes in the late summer of 2004, many homes were seriously damaged. Following the storms, the demand for repairs—including reroofing—far exceeded the supply of contractors who could provide such services. Meanwhile, many Floridians experienced power outages that left them without air conditioning, and suffered accordingly. Further, many homes sustained floor water damage—particularly in carpeted or wood floor areas.

Although coping with the aftermath of hurricanes is difficult, Florida Solar Energy Center (FSEC) looked to provide help by offering practical advice and applying research findings to those who were tackling repairs after a hurricane. Most homes can be made more energy efficient and comfortable, which is particularly advantageous in the event of power outages, and more resilient when facing future storms.

Cool Roofs

One major repair that must often be made after a hurricane is reroofing. By choosing the right roofing materials, you can make your entire home cooler, which will result in reduced energy use when your air conditioner is on, and more comfortable living conditions when power is not available.

Over the last decade, FSEC research has clearly shown that a white reflective tile or a white metal roof can reduce space-cooling loads by 20% or more. More recent testing has shown that unfinished Galvalume roofs—which look like brushed aluminum—can provide about half the cooling benefit of white roofs (see "Improving Attic Thermal Performance," *HE* Nov/Dec '04, p.12). Light-colored tile roofs can provide similar improvements in performance. FSEC tests show that attics covered by dark shingle roofs can reach temperatures as high as 140°F on summer days. Meanwhile, a white metal or white tile roof never gets much hotter than the outside temperature. Light-colored tile or metal roofs register somewhere between dark shingle roofs and white tile or metal roofs.

Though metal roofs can cost twice as much as shingle roofs, and tile roofs cost 3 times as much, they last at least 2 to 3 times as long and keep the home much cooler. That means average savings of about $120 per year in reduced air conditioning as well as a much cooler home when the power goes out and takes the air conditioning with it.

If you must choose shingles for structural reasons (many roofs cannot support the weight of tiles without physical strengthening), it is best to select the lightest-colored shingles available. You should also consider installing a foil radiant barrier on the underside of the roof decking. Testing performed for the Florida Power and Light Company showed that simply choosing white shingles would reduce space cooling by 4% , and choosing light-colored shingles with a radiant barrier should cut cooling costs by twice that or more. All these options will provide slightly better comfort on the interior when air conditioning is not available.

A standard, 2,500 ft^2 roof will cost \$5,000–\$6,000 to replace with shingles, \$8,000–\$12,000 to replace with metal, and about \$15,000 to replace with concrete barrel tile. Many homes cannot support the weight of a tile roof without extensive work beforehand, making metal roofs, such as factory-painted white or Galvalume, the more practical choice for many households.

A Stronger, More Forgiving Roof

When a house is exposed to hurricane forces, the roof is the building component most susceptible to damage, followed by the walls and openings, and then the foundation. Data collected by HUD studies after major hurricanes bear this out. These studies found that water penetration was also a major problem whenever roofing material was removed by wind action. For steep roof systems, many roofing failures occurred at the ridge or gable ends, where wind-induced forces were the highest. In a number of forensic evaluations, it was consistently found that the presence of gable ends increased the chances of roof failure after hurricanes.

Roofs that are likeliest to fail during hurricanes are poorly attached tile roofs or older lower-cost shingle roofs—particularly those applied with staples, which are now forbidden by current Florida building codes (the allowed fastener system is nails). Properly attached tile and metal roofs tend to do better in such storms, as evidenced in numerous post-storm studies conducted by HUD after Hurricanes Andrew, Hugo, and Iniki in the 1990s.

However, construction quality matters with all roofing systems. Properly installed hurricane straps, nails, and screws are a must. Tile roofs should be stoutly installed with screws rather than with mortar set, which is more prone to failure. Even with proper attachment for tile roofs, particular attention should be given to strong fasteners at the eaves, corners, and ridges where failure is more common. A properly attached shingle roof with high-quality architectural-grade shingles should do well.

In any case, homeowners seeking repairs should obtain multiple bids and deal only with licensed, reputable contractors. However, this is exceedingly difficult after a storm. Many homeowners with leaking roofs are looking for imme-

Better Glass for Coastal Homes

In the past fifteen years, research has shown that the leading cause of building failure during high-wind events is the failure of one or more windows or doors—which often fail due to flying debris, not because of the wind by itself. To minimize the possibility of window and door failure, impact-resistant glass products have been developed to withstand debris and wind, which in turn lowers structural vulnerabilities during a storm.

Impact-resistant windows use either laminated glass or shatter-resistant film. Shatter-resistant film is less expensive than laminated glass, and is often used in retrofitting operations. The film is placed on the interior of existing window glass, and upon impact, the film is designed to hold the shards of broken glass to the film, reducing injury to residents and home alike.

Laminated glass was developed for use in the automotive industry in the 1930s. Car windshields use laminated glass that has a $^{3}/_{100}$-of-an-inch plastic interlayer (usually polyvinyl butyral) in the middle of two regular pieces of glass. Researchers have found that by increasing this plastic interlayer to $^{9}/_{100}$-of-an-inch, laminated windows can provide better protection for homes during high-wind events. Even though the glass may break with a high velocity impact, the shattered glass adheres to the plastic interlayer, virtually eliminating glass hazards. When laminated glass windows are installed in a residence, they can usually withstand winds up to 130 miles per hour, including impacts by flying debris.

At a cost comparable to that of most high-quality windows, impact-resistant windows offer benefits including higher security, improved noise abatement, 99% protection from ultraviolet light, and increased energy efficiency.

Window companies are making even more improvements to laminated glass. For example, Hurd Millwork Company produces FeelSafe windows, which are Energy Star rated and also provide storm resistance that exceeds building code requirements for all coastal regions from Maine to Texas. Another company, CertainTeed, has released a new impact-resistant option for vinyl windows. These high-impact-resistant windows meet wind-borne debris testing standards of the International Building Code. The windows differ from other brands of laminated windows because they don't require special installation procedures.

Whether deciding on laminated windows or impact-resistant film, employing these technologies in homes will help to reduce damage during high winds. By providing impact resistant windows in new construction and retrofits, especially in tornado- and hurricane-prone climate areas, builders can add value to their product, which may set them apart from the competition. Whether you are a builder or a homeowner, using these window technologies will provide peace of mind before, during, and after the storm.

—Elka Karl is an associate editor at ***Home Energy*** *magazine.*

diate solutions, and that often means improvised fixes—such as tarpaulins. And the fact that older shingle roofs often fail when exposed to severe winds leaves many homeowners whose homes have shingle roofs looking for roofing contractors at the same time as they are recovering from the storm.

In spite of the energy and longevity benefits of the light-colored tile or metal roofs, many homeowners still choose shingles, partly because they are cheaper to install. Many homeowners choose dark-colored shingles for aesthetic reasons. Wind-resistant shingles are available, although they are more expensive. It is generally best to use three-tab self-sealing shingles with a six-nail application pattern, but you can choose light-colored shingles at no extra cost. Also, adding good ridge vents to a roofing system is another smart idea. The HUD post-hurricane research found that effective ridge ventilation was important in preventing sheathing uplift during hurricanes. Finally, if you install another shingle roof and have good attic ventilation, consider adding an attic radiant barrier to reduce attic heat gain.

In short, it is possible to improve a shingle roof—but a tile or metal roof will last longer and will better resist storm damage. Indeed, there are many metal roofs in central Florida that were installed in the early 1900s and are still providing reliable service. Some tile roofs in South Florida are still in use as well, even though they were installed in the 1950s. If you choose a cooler roofing system, such as light colored tile or metal, that is stoutly installed, you will be less likely to suffer storm damage and you will enjoy year-round savings on air conditioning.

HUD research also recommends that you consider the following issues to make your roof stronger:

• If you are installing a new roof with trusses, consider a hipped roof rather than one with gable ends. The latter tend to be more prone to storm damage. Highly pitched roofs with gable ends are particularly vulnerable. This is also the time to minimize roof penetrations, such as those for kitchen and bath exhaust vents, by running them out through the soffit. Note that furnace and range exhaust must still be run through the roof, however.

• You may also want to consider the new technique of unvented attics. Most Florida homes have vented attics, with batt or blown-in insulation applied just above the ceiling. This exposes air conditioning ductwork to very high temperatures and further magnifies any problems associated with duct leakage. Sealing the attic envelope and insulating at the roof deck provides a semiconditioned space for the ductwork, reducing conductive heat gain and minimizing the detrimental impact of any duct leakage. Water intrusion and hurricane resistance may also be improved.

If the building has gable ends, plan to have the ends stoutly braced. Do not install gable vents, which are prone to water leaks during wind-driven rain.

Plan to limit overhang width and instead use awnings, Bahama shutters, or other means to shade windows. Although porches are wonderful, they should be stoutly anchored.

• If the attic has soffit ventilation, it is important to provide ridge venting to prevent dangerous uplift on roof sheathing during high winds.

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Most Floridians know that it's important to use window hurricane shutters, such as the awning shutters shown in the photo, as a safety measure during storms.

• Verify that hurricane straps, nails (with shingles), and screws (with tile) are properly used.

• Renail any sheathing that looks loose.

• Be sure to get all of the roof vents and caps examined to see that they are properly attached and replace any that are weathered or damaged.

Windows and Doors

Most Floridians know that it's important to use hurricane shutters for windows as a safety measure during storms, but most Floridians have not considered how some systems—such as louvered Bahama shutters or awning shutters—can provide shading for windows when they are not being used to protect them from storms. Such shading devices can also be deployed more rapidly and easily than stored plywood or storm panels. And some of the new impact-resistant glass systems come with double-strength laminated glass, with solar-control properties that can cut down on heat transmission—a desirable characteristic nearly year-round in Florida's climate (see "Better Glass for Coastal Homes").

It is also useful to improve the anchorage of windows to openings that are old or in poor repair—particularly for a large expanse of glass such as a sliding glass door. Also, garage doors—particularly double-wide garage doors—can be strengthened to improve storm resistance.

Tile Floors

With the recent hurricanes, many homeowners have suffered water damage in the form of soaked carpets. Without power to dry them, there is an increased risk of mold or mildew, and many of these carpets have been ruined. Back during the 1960s, before most houses had air conditioning, and at a time when hurricanes in Florida were more prevalent, many Florida homes had tiled or terrazzo floors. Not surprisingly, a tiled or terrazzo floor is much less susceptible to water damage, and cleanup is simple. There are also no carpets to create a potentially devastating mold problem.

Perhaps just as important, FSEC research has shown that tiled floors feel cooler than carpets because they are in direct contact with the cooler ground underneath (the deep ground temperature in any location is equal to the average annual air temperature; for example, it's 72°F in Orlando). This free cooling is worth up to ½ a ton of air conditioning during early summer. And it can be a godsend when there is no air conditioning after a hurricane.

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Even more importantly, research, including that by Subrato Chandra at FSEC, has shown that, in Florida homes, carpeted floors can provide an attractive breeding ground for dust mites—a well-known source of allergens for many people (see "Allergy Relief in Humid Climates," *HE* Mar/Apr '02, p. 30).

All these factors suggest that replacing most of your moisture-damaged carpeted floors with tile will improve cooling and comfort and reduce allergic reactions in the occupants. Confine carpet to bedrooms only, or consider the use of throw rugs.

Although many homeowners will want to accomplish the repairs to their homes as soon as possible, they will do well to consider the long-term benefits of making energy-efficient choices as they make those repairs. Smart choices in building materials will provide superior protection against further storms while saving more energy. Equally important, replacing the damaged portions of your home with smart, energy-efficient materials will also ensure the comfort and health of your home.

Danny Parker is principal research scientist at the Florida Solar Energy Center. Parker has spent the last 26 years of his career in the field of energy efficiency research, and the last 16 at FSEC.

For more information:

The Disaster Contractors Network Web site (www.dcnonline.org) has detailed information on roof repairs, model contracts, emergency loans, and even a list of licensed Florida contractors. Visit this Web site for a great deal of important information that can help you rebuild after a hurricane or other disaster.

Check out the Florida Solar Energy Center's Web site (http://fsec.ucf.edu/pubs/designnotes/index.htm) for detailed information on radiant barrier installation.

For a case study on a home with an unvented attic, go to www.fsec.ucf.edu/bldg/baihp/data/CFRes/index.htm.

Storm Resistant Window and Door Manufacturers

Andersen Windows and Doors
Andersen Corporation
Bayport, MN 55003-1096
Tel: (800)426-4261
Web site:
www.andersenwindows.com

Hurd Windows and Doors
575 South Whelen Ave.
Medford, WI 54451
Tel: (800)223-4873
Web site: www.hurd.com

Loewen Windows
77 Hwy. 52 West
Box 2260
Steinbach, Manitoba R5G 1B2
Tel: (800)563-9367
Web site: www.loewen.com

Marvin Windows and Doors
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Tel: (888)537-7828
(United States)
Tel: (800)263-6161 (Canada)
Web site: www.marvin.com

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102 Main St.
Pella, IA 50219
Tel: (888)847-3552
Web site: www.pella.com

PGT Industries
Tel: (877)550-6006
Web site: www.winguard.com

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Parkersburg, WV 26102-1646
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Web site: www.simonton.com

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71 Progress Ave.
Cranberry Township, PA
16066
Tel: (800)837-7003
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One Weather Shield Plaza
P.O. Box 309
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